Math 107 Midterm 2 practice

Last updated: October 8, 2026.

Based on Lectures 18–29 and Quizzes 5–9. Use with the study guide. Exam: Midterm 2 · Friday, November 13.

Three types of problems will appear on Midterm 2: short answer definitions and meaning, computational and applied problems.

A. Definitions and meaning

Write a short precise answer (one or two sentences, or a formula). No long essays.

A1. What does it mean to say that A is an $m \times n$ matrix? What size is each column of A?

A2. Give the **column view** formula for Ax when A has columns $a_1, \dots, a_n$ and $x = (x_1, \dots, x_n)$.

A3. In one sentence: what does it mean to interpret A as a **transformation** from $\mathbb{R}^n$ to $\mathbb{R}^m$?

A4. Name three structured matrix types from class (beyond "square") and one practical consequence of structure for at least one of them.

A5. Translate "$Ax = b$ has a solution" into a statement about the **column space** of A.

A6. What is **reduced row echelon form** (RREF)? What are **pivot** and **free** variables?

A7. When does $Ax = b$ have (i) no solution, (ii) a unique solution, (iii) infinitely many solutions? Answer in terms of RREF / consistency / free variables for a square or rectangular system.

A8. Write the model for a **linear dynamical system** used in class. What do the state x_t and matrix A represent?

A9. When is the product AB defined? What are the dimensions of AB if A is $m \times n$ and B is $n \times p$?

A10. Why is matrix multiplication **not** generally commutative? Give a one-line reason (sizes or a counterexample idea).

A11. How does $A^k x$ relate to the dynamical system $x_{t+1} = Ax_t$?

A12. What does it mean for A to be **invertible**? How does invertibility relate to uniqueness of solutions of $Ax = b$?

A13. Why is solving $Ax = b$ generally preferable to forming A^{-1} and computing $A^{-1}b$?

A14. Define the **column space** $C(A)$ and the **null space** $N(A)$.

A15. Define $\operatorname{rank}(A)$. How does rank relate to $\dim C(A)$ and $\dim N(A)$ for an $m \times n$ matrix?

A16. For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, write $\det A$. What does $|\det A|$ mean geometrically for the image of the unit square?

A17. True or false, with a one-line reason: "If $\det A = 0$, then A is invertible."

A18. True or false, with a one-line reason: "If $AB = I$, then BA must also equal I for square A, B of the same size." (You may assume the course's usual finite-dimensional setting.)

A19. Light review (Quiz 5): What is a **basis** of a subspace V? What is $\dim V$?

A20. For each expression, say whether the result is a **scalar**, an n**-vector**, an $m \times n$ **matrix**, a **set of vectors**, or **undefined** (brief reason if undefined). Assume sizes make sense unless noted.

(a) Ax when A is $m \times n$ and x is an n-vector

(b) $A + B$ when A and B are both $m \times n$

(c) AB when A is 2×3 and B is 2×3

(d) A^{T} when A is $m \times n$

(e) $\det A$ when A is 2×2

(f) A^{-1} when A is singular 2×2

(g) $C(A)$

(h) $N(A)$

(i) $\operatorname{rank}(A)$

(j) A^2 when A is 3×2

(k) ABx when A is 2×2, B is 2×2, and x is a 2-vector

(l) residual $Ax - b$ when $Ax = b$ is consistent

B. Computational

Show your work for each calculation.

Matrices, Ax, and transforms

B1. Let

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Compute Ax using the column-combination view.

B2. Same A as B1. What are the input and output dimensions of the map $x \mapsto Ax$?

B3. Let

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Compute $R \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $R \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Describe the geometric effect of R on $\mathbb{R}^2$.

B4. Let

$$S = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Compute $S \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $S \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. What kind of transform is this (scaling / reflection / shear / rotation)?

B5. Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Compute $A + B$, $2A$, and A^{T}.

B6. Is

$$T = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

symmetric? Triangular? Diagonal?

Systems, elimination, RREF

B7. Write the linear system

$$x + 2y = 5,$$
$$3x - y = 1$$

as $Ax = b$. Give A, x, and b.

B8. Solve the system in B7 by hand (elimination or substitution). How many solutions?

B9. The augmented matrix $[A \mid b]$ row-reduces to

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Identify pivot variables and free variables. Write the general solution parametrically.

B10. Same RREF idea, but the last row is [0 0 0 | 4]. How many solutions? Why?

B11. Reduce

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 6 \end{array}\right]$$

to RREF. Describe the solution set of the corresponding system.

B12. For

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 7 \end{bmatrix},$$

is b in $C(A)$? Brief reason (no need for full RREF if you see the issue quickly).

Dynamics

B13. Let

$$A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Compute $x_1 = Ax_0$ and $x_2 = Ax_1$.

B14. A two-state system has transition matrix

$$A = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$$

(urban/rural fractions as in lecture). If $x_0 = (0.6, 0.4)$, compute $x_1 = Ax_0$ and interpret the entries.

B15. For the system $x_{t+1} = Ax_t$ with the A from B13, express x_3 in terms of A and x_0 using a matrix power (do not expand the power by hand unless you want to check).

Matrix products, powers, cost

B16. Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Compute AB and BA. Are they equal?

B17. With A from B16, compute A^2.

B18. Dimensions: A is 4×3, B is 3×2, C is 2×5. Is $(AB)C$ defined? What size is it? Is $A(BC)$ defined?

B19. Suppose a is a 1000-vector, b is a 1000-vector, and c is a 1000-vector. Compare forming $(ab^{\mathsf{T}})c$ versus $a(b^{\mathsf{T}}c)$ in words: which forms a huge intermediate matrix, and which is cheaper?

B20. Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Compute A^2 and A^3. Guess a formula for A^k.

Inverses

B21. Find the inverse of

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

(if it exists) by finding a matrix B with $AB = I$.

B22. Is

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

invertible? Give two different reasons (e.g. determinant, dependent columns, or null space).

B23. Solve

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

without forming an inverse explicitly (back substitution / elimination is fine).

Rank, column space, null space

B24. Let

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Describe $C(A)$. Is $b = (1, 2, 3)$ in $C(A)$? Is $b = (1, 3, 0)$ in $C(A)$?

B25. Same A as B24. Find $N(A)$ (solve $Ax = 0$). Give a basis for $N(A)$, or state that only 0 is in the null space.

B26. The RREF of a 3×4 matrix A is

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

What is $\operatorname{rank}(A)$? Which original columns form a basis for $C(A)$ (pivot columns)? Find a basis for $N(A)$.

B27. For the matrix in B26, what is $\dim N(A)$? Check that $\operatorname{rank}(A) + \dim N(A) = n$.

Determinants

B28. Compute

$$\det \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}.$$

B29. Compute

$$\det \begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 4 \\ 0 & 0 & 5 \end{bmatrix}.$$

B30. The linear map given by $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ sends the unit square to a rectangle. What is the area of that rectangle? How does this relate to $\det A$?

B31. Does

$$\det \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = 0$$

tell you that the columns are dependent? That A is noninvertible? That $Ax = b$ fails for some b?

C. Applied / harder

Show reasoning. These mix ideas or need a careful choice of tool.

C1. Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

(a) Solve $Ax = b$.
(b) Interpret A as a transformation of $\mathbb{R}^2$: what does it do to the standard basis vectors?
(c) Why does this system have a unique solution?

C2. A café stores each day as a 3-vector (coffee sold, pastry sold, total items), where total = coffee + pastry.

(a) Write a matrix A (or describe its columns) so that "feature vector" $= Ax$ for an underlying 2-vector $x = (\text{coffee}, \text{pastry})$.
(b) Are the three feature directions dependent? How does that show up in $\text{rank}(A)$?
(c) Why might this hurt a method that treats the three features as independent coordinates?

C3. Dynamics and products. Let $x_{t+1} = Ax_t$ with

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

(a) Compute x_1, x_2, and x_3.
(b) Compute A^2. How does that explain x_2?
(c) In one sentence: what long-run behavior do you see?

C4. Composition. A point is first reflected across the y-axis by

$$F = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix},$$

then rotated 90° counterclockwise by

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

(a) Which product, RF or FR, applies reflection first, then rotation? (Remember: $x \mapsto Fx$ then $x \mapsto Rx$ means $R(Fx) = (RF)x$.)
(b) Compute that product and apply it to $(1, 0)$.

C5. Cost. You need $y = ab^{\mathsf{T}}c$ where $a, b, c \in \mathbb{R}^n$ and n is large.

(a) What is the size of the intermediate matrix ab^{T}?
(b) What scalar do you get from $b^{\mathsf{T}}c$ first, and how do you finish?
(c) Which order would you choose, and why?

C6. Rank and consistency. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}.$$

(a) Argue that the first two columns are dependent.

(b) Find rank(A) (you may reduce or reason about independent columns).
(c) For which of $b = (0, 0, 0)$, $b = (3, 6, 1)$, $b = (1, 0, 0)$ is $Ax = b$ guaranteed / possible / impossible? Brief reasons.

C7. Null space and dependence. Columns of A are dependent if and only if $N(A)$ contains a nonzero vector. Use that fact on

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix};$$

find a nonzero x with $Ax = 0$, and write the corresponding dependence relation among the columns.

C8. Determinants and invertibility. For each matrix, compute the determinant and say whether it is invertible:

(i) $\begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix}$

(iii) $\begin{bmatrix} 5 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

C9. Mixed "which tool?" — for each task, name the best primary tool from: Ax / column combination, geometric transform, RREF / free variables, dynamical iteration $A^k x$, matrix product / composition, computational reordering, inverse, rank / column space / null space, determinant.

Task	Tool
"Will this 2×2 squash area to zero?"	
"Is b a combination of the columns?"	
"Write all solutions when there is a free variable"	
"Undo a reversible transform"	
"Where is the population after 10 steps?"	
"Reflect, then rotate"	
"Avoid building a huge outer product"	
"How many independent columns?"	

C10. Software interpretation. A classmate runs an RREF command and gets

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

for $[A \mid b]$. They conclude "the only solution is $x = 0$." What did they miss? Give the correct general solution.

C11. Geometry of det. Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

(a) Sketch the images of $e_1 = (1, 0)$ and $e_2 = (0, 1)$ under A, and the parallelogram they span.
(b) Compute $\det A$. Does the parallelogram have area 1?
(c) Is A invertible? Relate your answer to the parallelogram not collapsing to a line.

C12. Basis refresh + column space. Let $V = C(A)$ for

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 1 & 2 & 3 \end{bmatrix}.$$

(a) Find a basis for V and $\dim V$.
(b) Is $(1, 0, 1)$ in V? Is $(1, 1, 1)$ in V?
(c) Find a basis for $N(A)$.

C13. Build a transition matrix. Two classrooms: each day, 80% of students in Room 1 stay, 20% move to Room 2; 70% in Room 2 stay, 30% move to Room 1. Let $x_t = (r_1, r_2)$ be the counts (or fractions) in the two rooms.

(a) Write the matrix A in $x_{t+1} = Ax_t$.
(b) If $x_0 = (100, 0)$, compute x_1.
(c) Does A preserve the total number of students? Check x_1 entries sum.

Answers

Check only after a closed-book attempt. Update your definitions/formulas list for every miss.

A — Definitions (key points)

A1. A has m rows and n columns; each column is an m-vector.

A2. $Ax = x_1a_1 + \cdots + x_na_n$.

A3. A maps each input $x \in \mathbb{R}^n$ to the output $Ax \in \mathbb{R}^m$.

A4. e.g. identity, diagonal, triangular, symmetric, sparse, zero — sparse saves storage/ops; triangular makes determinants/back-substitution easy; etc.

A5. $b \in C(A)$ (equivalently: b is a linear combination of the columns).

A6. RREF is a row-echelon form with pivots 1 and zeros above/below pivots. Pivot variables correspond to pivot columns; free variables are the non-pivot unknowns.

A7. (i) inconsistent row $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$; (ii) consistent and no free variables; (iii) consistent with at least one free variable.

A8. $x_{t+1} = Ax_t$; x_t is the state, A is the transition / update matrix.

A9. Inner dimensions match (A's columns = B's rows); AB is $m \times p$.

A10. Order matters: AB and BA need not both be defined, and even when both are, they need not be equal.

A11. $x_k = A^k x_0$ (after k steps).

A12. There exists A^{-1} with $AA^{-1} = A^{-1}A = I$. For square A, invertible iff $Ax = b$ has a unique solution for every b.

A13. Forming A^{-1} is more work / less stable; solving the system directly is the standard approach.

A14. $C(A) = \text{span}\{\text{columns of } A\}$; $N(A) = \{x : Ax = 0\}$.

A15. Rank = dimension of the column space (= number of pivots). For $m \times n$ matrix: $\dim C(A) = \text{rank}(A)$ and $\dim N(A) = n - \text{rank}(A)$.

A16. $ad - bc$; absolute value = area scaling factor for the unit square (signed area accounts for orientation).

A17. False — $\det A = 0$ means noninvertible.

A18. True (in this course's finite-dimensional square-matrix setting): a one-sided inverse is a two-sided inverse.

A19. Independent spanning list; dimension = number of vectors in any basis.

A20. (a) m**-vector** (b) $m \times n$ **matrix** (c) **undefined** — inner dimensions $3 \neq 2$ (d) $n \times m$ **matrix** (e) **scalar** (f) **undefined** — singular, no inverse (g) **set of vectors** (h) **set of vectors** (i) **scalar** (nonnegative integer) (j) **undefined** — A is not square / AA sizes fail (k) 2**-vector** (l) m**-vector** that is 0 when consistent (here the zero vector)

B — Easy computations

B1. $Ax = 2\begin{bmatrix}1\\0\end{bmatrix} + 1\begin{bmatrix}-1\\3\end{bmatrix} + (-1)\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}-1\\2\end{bmatrix}$.

B2. Input $\mathbb{R}^3$, output $\mathbb{R}^2$.

B3. $Re_1 = (0, 1)$, $Re_2 = (-1, 0)$; 90° counterclockwise rotation.

B4. $Se_1 = (1, 0)$, $Se_2 = (2, 1)$; **shear** (horizontal shear).

B5. $A + B = \begin{bmatrix}2 & 1\\1 & 3\end{bmatrix}$; $2A = \begin{bmatrix}4 & 0\\0 & 6\end{bmatrix}$; $A^\mathsf{T} = A$ (already symmetric diagonal).

B6. Symmetric: yes. Triangular: no. Diagonal: no.

B7. $A = \begin{bmatrix}1 & 2\\3 & -1\end{bmatrix}$, $x = \begin{bmatrix}x\\y\end{bmatrix}$, $b = \begin{bmatrix}5\\1\end{bmatrix}$.

B8. Unique solution: e.g. $x = 1$, $y = 2$ (check: $1 + 4 = 5$, $3 - 2 = 1$).

B9. Pivots x_1, x_2; free $x_3 = t$. General: $x = (3 - 2t,\ 1 + t,\ t)$.

B10. No solutions — inconsistent ($0 = 4$).

B11. RREF $\left[\begin{array}{cc|c}1 & 2 & 3\\0 & 0 & 0\end{array}\right]$; infinitely many: $x_1 = 3 - 2t$, $x_2 = t$.

B12. No — second row of A is 2× first, but $7 \neq 2 \cdot 3$.

B13. $x_1 = (0.5, 0.5)$; $x_2 = (0.5, 0.5)$.

B14. $x_1 = (0.9 \cdot 0.6 + 0.2 \cdot 0.4,\ 0.1 \cdot 0.6 + 0.8 \cdot 0.4) = (0.62, 0.38)$; urban / rural fractions tomorrow.

B15. $x_3 = A^3 x_0$.

B16. $AB = \begin{bmatrix}2 & 1\\1 & 0\end{bmatrix}$, $BA = \begin{bmatrix}0 & 1\\1 & 2\end{bmatrix}$; **not** equal.

B17. $A^2 = \begin{bmatrix}1 & 4\\0 & 1\end{bmatrix}$.

B18. Yes, size 4×5; yes, $A(BC)$ also 4×5.

B19. $(ab^\mathsf{T})c$ builds a 1000×1000 matrix; $a(b^\mathsf{T}c)$ only needs a dot product then a scale — much cheaper.

B20. $A^2 = \begin{bmatrix}1 & 2\\0 & 1\end{bmatrix}$, $A^3 = \begin{bmatrix}1 & 3\\0 & 1\end{bmatrix}$; $A^k = \begin{bmatrix}1 & k\\0 & 1\end{bmatrix}$.

B21. $B = \begin{bmatrix}1 & -2\\0 & 1\end{bmatrix}$ (check $AB = I$).

B22. Not invertible: $\det = 0$; columns dependent ($c_2 = 2c_1$); nontrivial null space, e.g. $(2, -1)$.

B23. From second row $x_2 = 2$; first row $x_1 + x_2 = 3$ so $x_1 = 1$.

B24. $C(A) = \text{span}\{(1, 2, 0), (0, 0, 1)\} = \{(x, 2x, z)\}$. $(1, 2, 3)$: **yes**; $(1, 3, 0)$: **no** (middle entry must be twice the first).

B25. $Ax = 0$ forces $x_1 = 0$, $x_2 = 0$; $N(A) = \{0\}$ (two pivot columns; only the trivial null vector).

B26. Rank 2; pivot columns 1 and 3 of the original A. Free variables $x_2 = s$, $x_4 = t$. From RREF: $x_1 + 2s + 3t = 0$, $x_3 - t = 0$, so

$$x = s\begin{bmatrix}-2\\1\\0\\0\end{bmatrix} + t\begin{bmatrix}-3\\0\\1\\1\end{bmatrix}.$$

Basis for $N(A)$: those two vectors.

B27. $\dim N(A) = 2$; $2 + 2 = 4 = n$.

B28. $12 - 2 = 10$.

B29. Product of diagonal: $2 \cdot (-3) \cdot 5 = -30$.

B30. Area 6; equals $|\det A| = 6$.

B31. Yes; yes; yes (column space is not all of $\mathbb{R}^2$).

C — Harder

C1. (a) $x_2 = 2$, $x_1 = 2$. (b) $e_1 \mapsto e_1$, $e_2 \mapsto (1, 1)$ — shear. (c) Square invertible matrix / unique RREF pivots in both columns / $\det A = 1 \neq 0$.

C2. (a) e.g. $A = \begin{bmatrix}1 & 0\\0 & 1\\1 & 1\end{bmatrix}$. (b) Yes — third row = sum of first two; $\text{rank}(A) = 2$. (c) Redundant feature double-counts / distorts distance-based methods.

C3. (a) $x_1 = (1, 0)$, $x_2 = (0, 0)$, $x_3 = (0, 0)$. (b) $A^2 = 0$, so $x_2 = A^2 x_0 = 0$. (c) Dies to zero in two steps.

C4. (a) RF. (b) $RF = \begin{bmatrix}0 & -1\\-1 & 0\end{bmatrix}$; $(RF)(1, 0) = (0, -1)$.

C5. (a) $n \times n$. (b) Scalar $b^\mathsf{T} c$, then multiply by a. (c) $a(b^\mathsf{T} c)$ — avoids the huge matrix.

C6. (a) Column $2 = 2\times$ column 1. (b) $\text{rank}(A) = 2$ (e.g. columns 1 and 3 are independent; column 2 is redundant). (c) $b = (0, 0, 0)$ always solvable; $b = (3, 6, 1)$ is column 3, so solvable; $b = (1, 0, 0)$ not in $C(A)$ (any combination has second entry equal to twice the first).

C7. e.g. $x = (2, -1)$; $2c_1 - c_2 = 0$.

C8. (i) 8, invertible; (ii) 0, not invertible; (iii) -15, invertible.

C9. Determinant; column space / $Ax = b$; RREF / free variables; inverse; dynamical iteration $A^k x$; matrix product / composition; computational reordering; rank / independent columns.

C10. Free variable $x_3 = t$; solution $x = (-2t,\ t,\ t)$, not only zero.

C11. (a) Images $(1, 0)$ and $(1, 1)$; parallelogram is the unit-area shear of the square. (b) $\det A = 1$; area 1. (c) Yes invertible — parallelogram has positive area / does not collapse.

C12. (a) Basis e.g. $\{(1, 0, 1)\}$; dim = 1. (b) $(1, 0, 1)$ yes; $(1, 1, 1)$ no (middle entry). (c) $x_1 + 2x_2 + 3x_3 = 0$; basis e.g. $\{(-2, 1, 0), (-3, 0, 1)\}$.

C13. (a) $A = \begin{bmatrix}0.8 & 0.3\\0.2 & 0.7\end{bmatrix}$ (columns: where each room's students go — column 1 from Room 1, column 2 from Room 2). (b) $x_1 = (80, 20)$. (c) Yes — total 100 preserved (columns sum to 1).